Template Week 4 – Software

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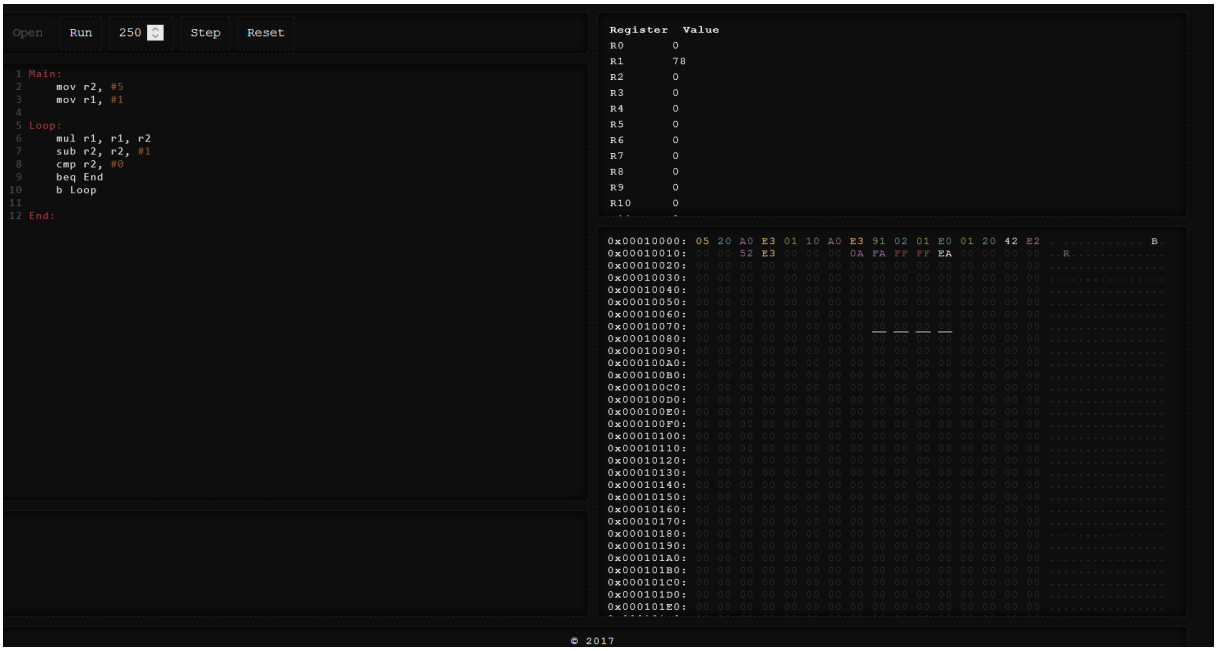
Assignment 4.1: ARM assembly

Screenshot of working assembly code of factorial calculation:



Assignment 4.2: Programming languages

Take screenshots that the following commands work:

javac --version

java --version

gcc --version

python3 --version

bash –version

```
tim@DellG15Tim:~$ javac --version
javac 21.0.9
tim@DellG15Tim:~$ java --version
openjdk 21.0.9 2025-10-21
OpenJDK Runtime Environment (build 21.0.9+10-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.9+10-Ubuntu-124.04, mixed mode, sharing)
tim@DellG15Tim:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

tim@DellG15Tim:~$ python3 --version
Python 3.12.3
tim@DellG15Tim:~$ bash --version
GNU bash, version 5.2.21(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
tim@DellG15Tim:~$
```

Assignment 4.3: Compile

Which of the above files need to be compiled before you can run them?

Fib.c

Fibonacci.java

Which source code files are compiled into machine code and then directly executable by a processor?

Fib.c

Which source code files are compiled to byte code?

Fibonacci.java

Which source code files are interpreted by an interpreter?

Fib.py

Fib.sh

These source code files will perform the same calculation after compilation/interpretation. Which one is expected to do the calculation the fastest?

Fib.c

How do I run a Java program?

Java Fibonacci.java of javac fibonacci en daarna java fibonacci

How do I run a Python program?

Type python (of python3) en daarna de naam van het programma dus in het geval van het zipje "python3 fib.py"

How do I run a C program?

Compile het met gcc en daarna kan je het als executable markeren met chmod +x "bestandsnaam". In het geval van fib eerst "gcc -o fib fib.c" en dan "chmod +x fib" als laatst kan je het draaien door "bash ./fib" of "./fib"

How do I run a Bash script?

bash ./scriptnaam

If I compile the above source code, will a new file be created? If so, which file?

Ja, de naam die je het specificeert, voor fib.c is dat fib

Take relevant screenshots of the following commands:

- Compile the source files where necessary
- Make them executable
- Run them
- Which (compiled) source code file performs the calculation the fastest?

```
tim@DellG15Tim:~/ITFundamentals/zipje/code$ ls -la
total 52
drwxr-xr-x 2 tim  tim   4096 Jan  8 17:01 .
drwxr-xr-x 4 tim  tim   4096 Dec  4 10:40 ..
-rw-r--r-- 1 tim  tim   1448 Jan  8 17:01 Fibonacci.class
-rw-r--r-- 1 tim  tim    839 Jun  9  2023 Fibonacci.java
-rwxr-xr-x 1 tim  tim  16136 Dec  4 09:26 fib
-rw-r--r-- 1 tim  tim    831 Jun  9  2023 fib.c
-rw-r--r-- 1 tim  tim    516 Jun  9  2023 fib.py
-rwxr-xr-x 1 tim  tim    668 Jun  9  2023 fib.sh
-rw-r--r-- 1 root root   100 Dec  4 10:07 macht.py
-rwxr-xr-x 1 tim  tim    249 Jun  9  2023 runall.sh
tim@DellG15Tim:~/ITFundamentals/zipje/code$
```

```
tim@DellG15Tim:~/ITFundamentals/zipje/code$ ls
Fibonacci.class  Fibonacci.java  fib  fib.c  fib.py  fib.sh  macht.py  runall.sh
tim@DellG15Tim:~/ITFundamentals/zipje/code$ ./fib
Fibonacci(18) = 2584
Execution time: 0.02 milliseconds
tim@DellG15Tim:~/ITFundamentals/zipje/code$ bash ./fib.sh
Fibonacci(18) = 2584
Excution time 3683 milliseconds
tim@DellG15Tim:~/ITFundamentals/zipje/code$ java Fibonacci
Fibonacci(18) = 2584
Execution time: 0.16 milliseconds
tim@DellG15Tim:~/ITFundamentals/zipje/code$ python3 fib.py
Fibonacci(18) = 2584
Execution time: 0.20 milliseconds
tim@DellG15Tim:~/ITFundamentals/zipje/code$
```

Assignment 4.4: Optimize

Take relevant screenshots of the following commands:

a) Figure out which parameters you need to pass to **the gcc** compiler so that the compiler performs a number of optimizations that will ensure that the compiled source code will run faster. **Tip!** The parameters are usually a letter followed by a number. Also read **page 191** of your book, but find a better optimization in the man pages. Please note that Linux is case sensitive.

```
Optimization Options
    -faggressive-loop-optimizations    -falign-functions[=n[:m:[n2[:m2]]]]    -falign-jumps[=n[:m:[n2[:m2]]]]
    -falign-labels[=n[:m:[n2[:m2]]]]    -falign-loops[=n[:m:[n2[:m2]]]]    -fno-allocation-dce
    -fallow-store-data-races   -fassociative-math    -fauto-profile    -fauto-profile[=path]    -fauto-inc-dec
    -fbranch-probabilities   -fcaller-saves   -fcombine-stack-adjustments    -fconserve-stack   -fcompare-elim
    -fcprop-registers    -fcrossjumping    -fcse-follow-jumps    -fcse-skip-blocks    -fcx-fortran-rules
    -fcx-limited-range  -fdata-sections   -fdce  -fdelayed-branch -fdelete-null-pointer-checks  -fdevirtualize
    -fdevirtualize-speculatively    -fdevirtualize-at-ltrans    -fdse    -fearly-inlining    -fipa-sra
    -fexpensive-optimizations    -ffat-lto-objects    -ffast-math    -ffinite-math-only    -ffloat-store
    -fexcess-precision=style  -ffinite-loops  -fforward-propagate  -ffp-contract=style   -ffunction-sections
    -fgcse   -fgcse-after-reload  -fgcse-las  -fgcse-lm  -fgraphite-identity -fgcse-sm  -fhoist-adjacent-loads
    -fif-conversion -fif-conversion2  -findirect-inlining  -finline-functions   -finline-functions-called-once
    -finline-limit=n  -finline-small-functions  -fipa-modref  -fipa-cp  -fipa-cp-clone -fipa-bit-cp  -fipa-vrp
    -fipa-pta    -fipa-profile    -fipa-pure-const    -fipa-reference    -fipa-reference-addressable
    -fipa-stack-alignment   -fipa-icf   -fira-algorithm=algorithm  -flive-patching=level  -fira-region=region
    -fira-hoist-pressure    -fira-loop-pressure    -fno-ira-share-save-slots    -fno-ira-share-spill-slots
    -fisolate-erroneous-paths-dereference    -fisolate-erroneous-paths-attribute    -fivopts
    -fkeep-inline-functions    -fkeep-static-functions    -fkeep-static-consts    -flimit-function-alignment
    -flive-range-shrinkage   -floop-block    -floop-interchange   -floop-strip-mine   -floop-unroll-and-jam
    -floop-nest-optimize    -floop-parallelize-all    -flra-remat    -flto    -flto-compression-level
    -flto-partition=alg    -fmerge-all-constants    -fmerge-constants    -fmodulo-sched
    -fmodulo-sched-allow-regmoves   -fmove-loop-invariants    -fmove-loop-stores    -fno-branch-count-reg
    -fno-defer-pop   -fno-fp-int-builtin-inexact  -fno-function-cse -fno-guess-branch-probability  -fno-inline
    -fno-math-errno    -fno-peephole   -fno-peephole2    -fno-printf-return-value    -fno-sched-interblock
    -fno-sched-spec  -fno-signed-zeros -fno-toplevel-reorder  -fno-trapping-math  -fno-zero-initialized-in-bss
    -fomit-frame-pointer   -foptimize-sibling-calls  -fpartial-inlining   -fpeel-loops  -fpredictive-commoning
    -fprefetch-loop-arrays -fprofile-correction -fprofile-use   -fprofile-use=path  -fprofile-partial-training
    -fprofile-values    -fprofile-reorder-functions    -freciprocal-math    -free    -frename-registers
    -freorder-blocks -freorder-blocks-algorithm=algorithm -freorder-blocks-and-partition   -freorder-functions
    -frerun-cse-after-loop   -freschedule-modulo-scheduled-loops  -frounding-math   -fsave-optimization-record
    -fsched2-use-superblocks    -fsched-pressure    -fsched-spec-load    -fsched-spec-load-dangerous
    -fsched-stalled-insns-dep[=n]    -fsched-stalled-insns[=n]    -fsched-group-heuristic
    -fsched-critical-path-heuristic    -fsched-spec-insn-heuristic    -fsched-rank-heuristic
    -fsched-last-insn-heuristic    -fsched-dep-count-heuristic    -fschedule-fusion    -fschedule-insns
    -fschedule-insns2    -fsection-anchors    -fselective-scheduling    -fselective-scheduling2
    -fsel-sched-pipelining    -fsel-sched-pipelining-outer-loops    -fsemantic-interposition    -fshrink-wrap
    -fshrink-wrap-separate    -fsignaling-nans    -fsingle-precision-constant    -fsplit-ivs-in-unroller
    -fsplit-loops  -fsplit-paths  -fsplit-wide-types   -fsplit-wide-types-early   -fssa-backprop  -fssa-phiopt
    -fstdarg-opt   -fstore-merging   -fstrict-aliasing   -fipa-strict-aliasing   -fthread-jumps    -ftracer
```

Je kan hier opties van gebruiken om het te optimaliseren. Het boek gebruikt de -O optie

b) Compile **fib.c** again with the optimization parameters

Gedaan met gcc -O -o fib fib.c

c) Run the newly compiled program. Is it true that it now performs the calculation faster?

Nee, hij is niet sneller

d) Edit the file **runall.sh**, so you can perform all four calculations in a row using this Bash script. So the (compiled/interpreted) C, Java, Python and Bash versions of Fibonacci one after the other.

Gedaan #!/bin/bash

clear

n=19

```
echo "Running C program:"
./fib $n
echo -e '\n'

echo "Running Java program:"
java Fibonacci $n
echo -e '\n'

echo "Running Python program:"
python3 fib.py $n
echo -e '\n'

echo "Running BASH Script"
./fib.sh $n
echo -e '\n'
```

Assignment 4.5: More ARM Assembly

Like the factorial example, you can also implement the calculation of a power of 2 in assembly. For example you want to calculate $2^4 = 16$. Use iteration to calculate the result. Store the result in r0.

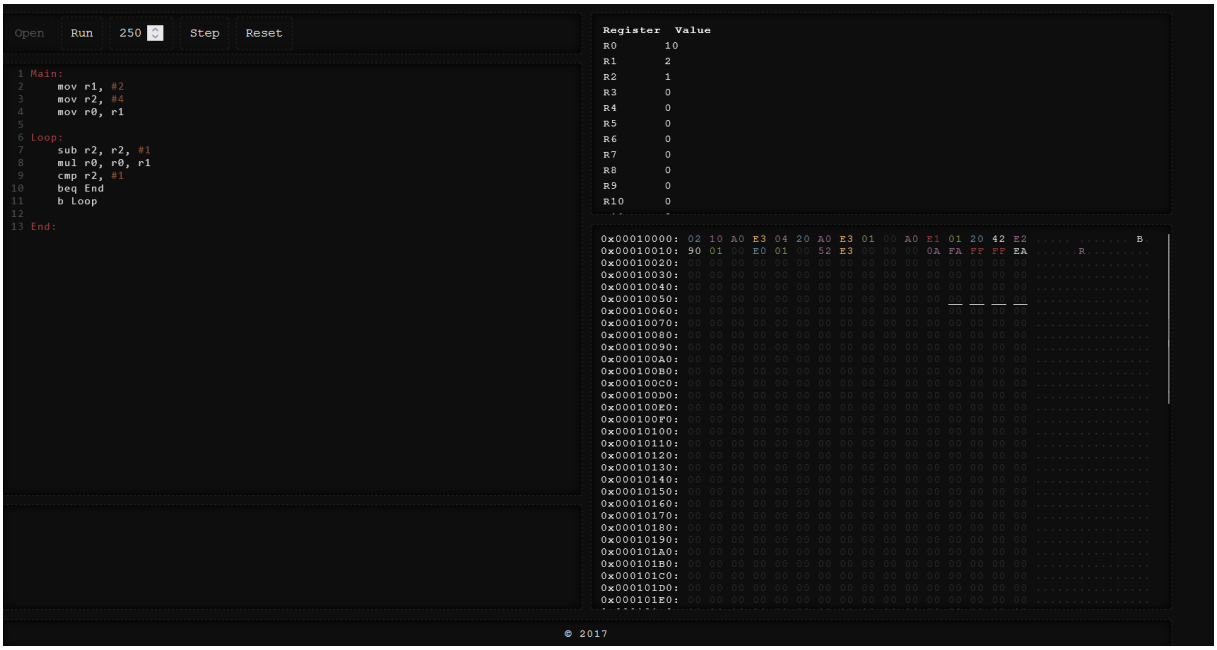
```
Main:
mov r1, #2
mov r2, #4

Loop:

End:
```

Complete the code. See the PowerPoint slides of week 4.

Screenshot of the completed code here.



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